

BankPrep

SBI PO Prelims - Reasoning & Arithmetic Study Pack

Part 1 - Seating Arrangement: 7 hard types, methods and solved puzzles

Part 2 - Seating practice sets A to G (unsolved)

Part 3 - Syllogism: method + 20 questions

Part 4 - Coding-Decoding: 6 types + 12 questions

Part 5 - Order & Ranking: formulas + 8 questions

Part 6 - Arithmetic: Profit & Loss, SI/CI, Time & Work, Partnership, Mixtures, Pipes & Cisterns

Part 7 - Answer keys for everything

Every method here was built and tested during the prep sessions. Work the questions on paper, then self-check against Part 7.

Exam: 1 August | English 40Q/20min | Quant 30Q/20min | Reasoning 30Q/20min

Negative marking: 0.25 per wrong answer. A rushed wrong answer swings 1.25 marks.

Attempt order: English, then Reasoning, then Quant. Skip any set that will not crack in about a minute.

Part 1 - Seating Arrangement: the 7 hard types

Every seating puzzle is the same skill: **turn words into a fixed frame, then place people by certainty** - locked clues first, floating clues last. What changes between types is only the frame and the left/right rule.

THE TWO COUNTING RULES - the most common source of lost marks

"Nth to the left/right of X" = move **exactly N seats**. 2nd = 2 seats over.

"K persons sit between X and Y" = they are **K+1 seats apart**.

Start counting at the **neighbour**, never at the person themselves. Say it out loud: "B one, D two, A three."

Type 1 - Linear with mixed facing (North / South)

Facing **NORTH** -> their right = **higher** position numbers.

Facing **SOUTH** -> their right = **lower** position numbers (flipped).

The direction is decided by the **reference person's** facing, not the subject's.

The impossibility trick: when a clue would push someone off the row, that impossibility *forces* the facing. If Y is at seat 7 of 7 and "C is 2nd to the right of Y", then Y facing North puts C at seat 9 - impossible. So Y faces South and C is at 5. You gain Y's direction for free.

Method: write positions 1-N with **L** and **R** marked at the ends. Test both facings on each direction clue; one usually dies instantly. Keep facings in a second row under the names.

Type 2 - Double row (parallel rows facing each other)

Two rows, equal numbers, facing each other. Person in column *i* of row 1 faces the person in column *i* of row 2.

The rows have **mirrored** left/right. The row facing **North** -> right = our right. The row facing **South** -> right = our left.

Bridge clue = "X faces Y" - it welds the two rows together by locking a column across both.

Draw them stacked with columns aligned and mark each row's arrow once. Solve one row fully, then use a "faces" clue to anchor the second.

Type 3 - Circular with mixed facing (some in, some out)

Facing **CENTRE** -> left = **clockwise**, right = anti-clockwise.

Facing **OUTSIDE** -> **flipped**: left = anti-clockwise, right = clockwise.

Opposite in a circle of 8 = **+4 seats**.

- Fix anyone at seat 1 and number seats 1-8 clockwise. Only relative positions matter, so this costs nothing.
- Write **I** or **O** beside every name the moment you learn a facing.
- The impossibility trick still works, but instead of falling off the row, a bad facing lands someone on an **already-occupied seat**.

Type 4 - Square / rectangular table

Eight people: four at the **corners** facing the **centre**, four at the **middle of the sides** facing **outside**. Some papers reverse it - always read which.

Why it is nasty: the facing alternates every seat, so left/right **zigzags** as you go around. Write the facing strip *once* before placing anybody, then every clue becomes a lookup:

seat	1	2	3	4	5	6	7	8
type	cor	mid	cor	mid	cor	mid	cor	mid
faces	IN	OUT	IN	OUT	IN	OUT	IN	OUT
left =	CW	ACW	CW	ACW	CW	ACW	CW	ACW

Type 5 - Uncertain count (unknown number of people)

You do not know how long the row is, so nothing anchors and everything floats. The escape is to hunt for **absolute anchors** first.

"Only K persons sit to the left of X" -> X is at seat **K+1**. Exact.

"X sits Nth from the left end" -> X = seat N. Exact.

Finding the total: "only K persons sit to the right of X" -> **total = position(X) + K**.

Conversion: position from right = **total - position from left + 1**.

Place the anchors, chain the relative clues off them, and **park any clue that floats** - it resolves later. Hard versions bury the anchor late in the clue list and tie the total to an attribute-holder rather than a named person.

Type 6 - Circular / linear with an attribute (compound)

Position plus facing plus attribute, all at once. The clean two-pass grid does not work here because the attribute clues *are* position clues.

Treat the attribute as a PLACEHOLDER PERSON. "The one who likes Blue sits opposite Q" -> write **Blue** on that seat like a name.

Keep placing placeholders alongside real names. When a later clue lands a **named person on a seat that already holds a placeholder**, that **collision** welds the attribute to the person.

Hunt for collisions - they are where the puzzle cracks open.

Two-pass grid still applies when position clues and attribute clues are separate (floor puzzles, month puzzles): solve **all positions first**, ignoring the attribute entirely, then drop attributes in via **bridge clues** that tie an attribute to a position.

Type 7 - Combo (seating + blood relation)

Two puzzles fused: a seating diagram *and* a family tree. The mistake is building both at once.

SPLIT every clue into two notes. "B, the wife of A, sits opposite A" becomes:

Position: B is opposite A. **Relation:** B = A's wife, B is female.

Pass 1 - seating only. Ignore every relation word.

Pass 2 - tree only. Now use the relation halves.

Mark **M/F** beside every name as gender is revealed. An unstated gender means **cannot be determined**.

Family terms - quick glossary

Term	Who they are	Term	Who they are
Nephew	brother's or sister's SON	Niece	brother's or sister's DAUGHTER
Uncle	parent's brother	Aunt	parent's sister
Cousin	uncle's or aunt's child	Sibling	brother or sister
Son-in-law	daughter's husband	Daughter-in-law	son's wife
Brother-in-law	spouse's brother / sister's husband	Sister-in-law	spouse's sister / brother's wife

Exam strategy for this block

- Puzzles and seating are **15 to 20 of the 30 reasoning marks** - the deciding block.
- A multi-layer monster (position + attribute + gender + opposite) is a **strategic skip**. If a set is not cracking in about a minute of reading, leave it and bank the easy sets.
- In multi-layer sets, attack the **attribute layer by elimination first** - it is often fully forced independent of the seats.
- Before writing any answer, say the walk out loud: "W faces out, so right is clockwise - seven, eight. V." Three seconds, and it converts build-accuracy into marks.

Part 2 - Seating practice sets (A to G)

Prelims difficulty. Sets E, F and G carry a second variable - those are the ones that decide the section. Answers in Part 7.

SET A - Linear, mixed facing (8 people)

Eight people sit in a row (positions 1-8, left to right). Some face **North**, some face **South**.

1. P sits **4th from the left** end.
2. **Two** persons sit between P and Q; Q is to the **right** of P.
3. R sits **2nd to the right** of Q.
4. S sits **3rd to the left** of R. **R faces the same direction as Q.**
5. T sits at an **extreme end**.
6. U sits **2nd to the right** of T.
7. V sits **3rd to the right** of U.
8. **Four** persons face South.
9. W and S face the **same** direction.
10. P and V face the **same** direction. **P faces North.**

- **A1.** Who sits at the extreme right end?
- **A2.** How many persons sit between W and R?
- **A3.** Who sits 3rd to the right of Q?
- **A4.** How many persons face North?
- **A5.** Who sits immediately to the left of V?

SET B - Circular + profession (8 people, all facing centre)

Eight people sit around a circular table, **all facing the centre**. Each has a different profession. H takes the one remaining seat.

1. A sits **3rd to the left** of D.
2. C sits **2nd to the right** of A.
3. B sits **opposite** C.
4. E sits **immediately to the right** of B.
5. F sits **3rd to the left** of E.
6. G sits **3rd to the left** of F.
7. **A is a Doctor.**
8. The **Doctor** sits **opposite** the **Teacher**.
9. The **Lawyer** sits **2nd to the right** of the Teacher.
10. The **Engineer** sits **immediately to the left** of the Doctor.

- **B1.** Who sits opposite D?
- **B2.** Who is the Teacher?
- **B3.** Counting **clockwise from B**, how many persons sit between B and C?
- **B4.** Who sits 3rd to the right of H?

- **B5.** Who is the Engineer?

SET C - Double row (12 people, 6 + 6)

Row 1 - P, Q, R, S, T, U - facing **South**. **Row 2** - A, B, C, D, E, F - facing **North**. Each row-1 person faces exactly one row-2 person.

1. P sits **3rd to the left** of T.
 2. T sits at an **extreme end** of row 1.
 3. Q sits **2nd to the right** of P.
 4. R sits **immediately to the left** of Q.
 5. S sits at an **extreme end** of row 1.
 6. **C faces P.**
 7. A sits **3rd to the left** of C.
 8. B sits **2nd to the right** of A.
 9. D sits **immediately to the right** of C.
 10. E does **not** sit at an extreme end.
- **C1.** Who faces R?
 - **C2.** Who sits at the two extreme ends of row 2?
 - **C3.** How many persons sit between C and F?
 - **C4.** Who sits 2nd to the left of the person who faces U?
 - **C5.** Who faces S?

SET D - Square table + attribute (8 people)

Eight people sit around a **square table**: four at the **corners** facing the **centre**, four at the **middle of the sides** facing **outside**. Q takes the one remaining seat.

1. J sits at a **corner**.
 2. The one who likes **Tea** sits **2nd to the right** of J.
 3. K sits **opposite** the one who likes Tea.
 4. L sits **3rd to the left** of K.
 5. M sits **immediately to the right** of L.
 6. N sits **2nd to the right** of M.
 7. O sits **immediately to the right** of N.
 8. P sits **2nd to the left** of O.
- **D1.** Who likes Tea?
 - **D2.** Who sits opposite O?
 - **D3.** Who sits 2nd to the left of J?
 - **D4.** Counting **clockwise from P**, how many persons sit between P and N?
 - **D5.** Who sits immediately to the right of Q?

SET E - Circular + mixed facing + month (double variable)

Eight people sit around a circular table; some face the **centre**, some face **outside**. Each was born in a different month.

1. A sits **3rd to the left** of B. **A faces the centre**.
 2. The one born in **March** sits **opposite** B.
 3. C sits **2nd to the right** of the one born in March. **The March-born faces outside**.
 4. D sits **immediately to the left** of C. **C faces the centre**.
 5. The one born in **July** sits **4th to the right** of C.
 6. **E is born in July**.
 7. F sits **2nd to the left** of E. **E faces the centre**.
 8. G sits **immediately to the left** of F.
 9. **Three** persons face outside.
 10. D and B face the **same** direction; G and H face the **same** direction. **D faces outside**.
 11. The one born in **January** sits **2nd to the right** of E.
 12. **H is born in September**.
 13. The one born in **April** sits **opposite** the one born in September.
 14. **C is born in December. D is born in February**.
- **E1.** Who is born in March?
 - **E2.** Who sits 3rd to the right of the one born in January?
 - **E3.** Counting **clockwise from D**, how many persons sit between D and H?
 - **E4.** Who sits opposite the one born in December?
 - **E5.** How many persons face the centre?

SET F - Floor + Flat grid (2D) + colour

Ten persons live in a **5-floor building with 2 flats per floor** - **Flat-1** (west) and **Flat-2** (east). Floors are numbered **1 (bottom) to 5 (top)**. J takes the one remaining flat.

1. **Two** persons live between A and E, and both live in **Flat-1**.
 2. E lives on the **lowest** floor.
 3. G lives **immediately below A**, in the same flat.
 4. B lives **immediately below G**, in the same flat.
 5. D lives in the **same flat as B**, on the **topmost** floor.
 6. C lives in **Flat-2**, on the **same floor as G**.
 7. F lives **immediately above C**, in the same flat.
 8. I lives on the **topmost** floor in Flat-2.
 9. H lives on the **lowest** floor in Flat-2.
 10. The one who likes **Red** lives on the 3rd floor in Flat-2.
 11. The one who likes **Blue** lives **immediately above** the Red-liker, in the same flat.
 12. The one who likes **White** lives on the **topmost floor in Flat-1**.
- **F1.** Who lives on the 2nd floor in Flat-2?
 - **F2.** How many persons live between A and B?
 - **F3.** Who likes Blue?

- **F4.** Who lives immediately above H?
- **F5.** Who likes White?

Treat this as a **table with two columns**, not a line: floors down the side, Flat-1 and Flat-2 as columns.

SET G - Double row + sport (attribute across both rows)

Row 1 - A, B, C, D, E - facing **South**. **Row 2** - P, Q, R, S, T - facing **North**. Each likes a different sport.

1. C sits **3rd to the right** of A.
 2. A sits at an **extreme end** of row 1.
 3. B sits **immediately to the left** of C.
 4. D sits at an **extreme end** of row 1.
 5. **R faces B.**
 6. P sits **2nd to the left** of R.
 7. Q sits **immediately to the right** of R.
 8. S does **not** sit at an extreme end.
 9. The one who likes **Cricket** faces D.
 10. The one who likes **Tennis** sits **2nd to the right** of the Cricket-liker.
 11. **E likes Hockey.**
 12. The one who likes **Football** sits **immediately to the left** of the Hockey-liker, in the same row.
- **G1.** Who faces C?
 - **G2.** Who likes Tennis?
 - **G3.** Who sits at the two extreme ends of row 1?
 - **G4.** Who sits 2nd to the left of Q?
 - **G5.** Who likes Football?

Part 2B - Session puzzles, worked (1 to 7)

These are the puzzles used to learn each type, with the solved arrangement underneath. Re-solve them cold in a week and check against the grid.

Puzzle 1 - Linear, mixed facing

Seven people A-G in a row (positions 1-7). Some face North, some face South. *A is 3rd from the left. Three persons sit between A and B. C is 2nd to the right of B. D is 3rd to the left of C. E is immediately to the right of D. G is 3rd to the right of E. Three persons face South. F and G face opposite directions. Both immediate neighbours of A face the same direction.*

Pos	1	2	3	4	5	6	7
Person	E	D	A	G	C	F	B
Faces	N	S	N	S	N	N	S

Key move: B is forced to seat 7; then "C is 2nd to the right of B" would put C at seat 9 if B faced North - impossible, so B faces South. The cascade runs from there.

Puzzle 2 - Double row

Row 1 (P, Q, R, S, T) faces South; Row 2 (A, B, C, D, E) faces North. *P is 3rd to the right of R. R is at an extreme end. S is 2nd to the left of P. Q is at an extreme end. A faces T. B is 2nd to the right of A. D is 3rd to the left of B. E is not at an extreme end.*

Column	1	2	3	4	5
Row 2 (North)	C	D	A	E	B
Row 1 (South)	Q	P	T	S	R

Answers: faces P = D; ends of row 1 = Q and R; immediately right of the one facing S = B; between D and B = 2; faces Q = C.

Puzzle 3 - Circular, mixed facing

Eight people A-H around a circle; some face the centre, some outside. *A is 2nd to the left of B (B faces centre). C is opposite A. D is 3rd to the right of C (C faces outside). E is 2nd to the right of D (D faces centre). F is opposite E. G is immediately to the right of F. Both neighbours of G face the same direction. E and H face opposite directions. Four face outside. A faces the same direction as B.*

Seat	1	2	3	4	5	6	7	8
Person	B	D	A	F	G	H	C	E
Faces	IN	IN	IN	OUT	OUT	OUT	OUT	IN

Key move: clue 6 - if F faced the centre, G would land on seat 3 which is already A. So F faces outside. That is the circular version of the impossibility trick.

Puzzle 4 - Square table

Eight people P-W: corners face the centre, mid-sides face outside. *P is at a corner. T is immediately to the left of P. R is 3rd to the right of T. Q is 2nd to the left of R. S is opposite Q. V is 3rd to the right of S. W is 2nd to the left of V.*

Seat	1	2	3	4	5	6	7	8
Type	cor	mid	cor	mid	cor	mid	cor	mid
Person	P	T	S	U	R	W	Q	V

Answers: opposite P = R; 2nd to the right of W = V; between U and Q clockwise = 2; immediately left of S = U; T with respect to R = 3rd to the right.

Puzzle 5 - Uncertain count

Only five persons sit to the left of P. Q is 4th to the right of P. Three persons sit between Q and R. S is 2nd to the left of R. T is exactly midway between P and Q. Only two persons sit to the right of R. U is 3rd to the left of T. V is immediately to the left of U.

Positions: V = 4, U = 5, P = 6, T = 8, Q = 10, S = 12, R = 14. **Total = 14 + 2 = 16.**

Answers: 16 in the row; between T and S = 3; Q is 7th from the right; 2nd to the right of V = P; between U and Q = 4.

Puzzle 5B - Uncertain count, hard (with an attribute)

Only three persons sit to the left of A. The Guava-liker is 6th to the right of A. B is 2nd to the left of the Guava-liker. Three persons sit between B and C. Only two persons sit to the right of C. The Litchi-liker is 3rd to the left of C. G is immediately to the left of the Guava-liker. D likes Guava. E is 4th to the left of D. F is exactly midway between E and B. The Papaya-liker is 3rd from the right end.

Positions: A = 4, E = 6, F = 7, B = 8, G = 9 (Litchi), D = 10 (Guava), C = 12 (Papaya). **Total = 12 + 2 = 14.**

Answers: 14 in the row; Litchi = G; E is 9th from the right; between F and D = 2; 3rd to the right of A = F.

Key move: clues 6 and 7 point at the same seat - that cross-link is how G gets a fruit without any clue naming it.

Puzzle 6 - Circular + attribute (compound)

Eight friends P-W, some facing centre, some outside, each liking a different colour. Q is 3rd to the left of P (P faces centre). The Blue-liker is opposite Q. R is 2nd to the right of the Blue-liker (who faces outside). S is immediately to the left of R (R faces centre). The Green-liker is 3rd to the right of S (S faces outside). T likes Green. U is 2nd to the left of T (T faces centre). V is immediately to the left of U. Four face outside. Q and V face the same direction. W faces the same direction as T.

Seat	1	2	3	4	5	6	7	8
Person	P	R	S	Q	W	T	V	U
Faces	IN	IN	OUT	OUT	IN	IN	OUT	OUT
Colour	-	-	-	-	-	Green	-	Blue

The collision: the Blue placeholder goes on seat 8 early; much later U lands on seat 8, which is what makes U the Blue-liker.

Puzzle 7 - Combo (seating + blood relation)

Eight family members around a circular table, all facing the centre. A is 2nd to the left of his daughter-in-law D. F sits between A and D. B, the wife of A, sits opposite A. C, the only son of A, is 2nd to the left of B. E is immediately to the left of C. H, the son of E, sits opposite E. G is immediately to the right of C. F and G are the children of C and D. E is the daughter of B.

Seat	1	2	3	4	5	6	7	8
Person	A	F	D	H	B	G	C	E

Family tree: A (M) is married to B (F). Their children are C (M) and E (F). D (F) is C's wife. F and G are the children of C and D. H (M) is E's son.

Answers: H is C's **nephew** (E is C's sister); opposite G = **F**; D is E's **sister-in-law**; 3rd to the left of B = **E**; clockwise between H and C = **2**.

Part 3 - Syllogism

THE ALGORITHM - run it in this order, every time

1. Is conclusion I forced? Is II forced? Both -> **(e)**. Exactly one -> **(a)** or **(b)**.
2. Neither forced -> are I and II an **exact opposite pair**? Yes -> **(c) Either/Or**. No -> **(d) Neither**.

The only two complementary pairs: **Some <-> No**, and **All <-> Some-not**.

Why the pair test works: "Some X are Y" means at least one; "No X is Y" means zero. Between them they cover every case with no gap, so one must be true even when you cannot tell which. "Some X are Y" vs "All X are Y" are both positive claims - they leave a gap, so that is Neither.

The 6 forced pairs - memorise these

Statements	Forced conclusion
All + All	All A are C
All + No	No A is C
Some + All	Some A are C
Some + No	Some A are not C
No + All	Some C are not A
No + Some	Some C are not A

Special case: All + Some -> Either/Or ("Some A are C" / "No A is C").

Free conversions: All A are B -> **Some B are A**. No A is B -> **No B is A**. Missing these turns a correct "Both" into "Only I".

"Only a few X are Y" = Some X are Y **and** Some X are not Y.

"All A are C" implies **"Some A are C"** - All always contains Some.

A flat conclusion must be **forced**. Check *possibility* only when the conclusion literally says "is a possibility".

Practice - 20 questions

Options for all: **(a)** Only I **(b)** Only II **(c)** Either I or II **(d)** Neither I nor II **(e)** Both I and II. Target: under 12 minutes for all 20.

1. All pens are books. All books are papers.
I: All pens are papers | II: Some papers are pens
2. All cats are dogs. No dog is a rat.
I: No cat is a rat | II: Some cats are rats
3. Some tables are chairs. All chairs are wooden.
I: Some tables are wooden | II: All tables are wooden
4. Some rings are gold. No gold is silver.
I: Some rings are not silver | II: All rings are silver
5. No apple is a mango. All mangoes are fruits.
I: Some fruits are not apples | II: No fruit is an apple

6. All roads are paths. Some paths are lanes.
I: Some roads are lanes | II: No road is a lane
7. All keys are locks. Some locks are doors.
I: All keys are doors | II: Some keys are not doors
8. Some birds are crows. Some crows are black.
I: Some birds are black | II: Some black are birds
9. Only a few pens are red. All red are bright.
I: Some pens are bright | II: All pens are bright
10. All flowers are roses. No rose is a lily.
I: No flower is a lily | II: Some lilies are flowers
11. Some books are novels. All novels are stories.
I: Some books are stories | II: Some stories are books
12. No stone is a rock. Some rocks are hard.
I: Some hard things are not stones | II: All hard things are stones
13. All cups are glasses. All glasses are plates.
I: All cups are plates | II: All plates are cups
14. Some pens are pencils. No pencil is an eraser.
I: Some pens are not erasers | II: All pens are erasers
15. All doctors are engineers. Some engineers are teachers.
I: All doctors being teachers is a possibility | II: Some teachers are doctors
16. Some fruits are vegetables. All vegetables are green.
I: All fruits being green is a possibility | II: Some fruits are green
17. All A are B. All B are C. Some C are D.
I: Some A are D | II: No A is D
18. Some P are Q. All Q are R. No R is S.
I: Some P are not S | II: All P are S
19. Only a few students are teachers. All teachers are graduates.
I: Some students are graduates | II: Some students are not graduates
20. No car is a bus. All buses are vehicles. Some vehicles are trucks.
I: Some vehicles are not cars | II: All trucks being cars is a possibility

Part 4 - Coding-Decoding

Type 1 - Letter shift (Caesar)

Every letter moves by a fixed amount. Find the shift from one letter, confirm on a second, then apply.

- **Ex 1:** CAT -> DBU. C to D is **+1**, A to B is **+1**. So **DOG -> EPH**.
- **Ex 2:** MOUSE -> LNTRD. M to L is **-1**. So **TIGER -> SHFDQ**.
- **Growing shift variant:** FRIEND -> GTLISJ. F+1=G, R+2=T, I+3=L, E+4=I, N+5=S, D+6=J.

Type 2 - Opposite letters (positions sum to 27)

A-Z, B-Y, C-X and so on. Landmarks **EJOTY**: E=5, J=10, O=15, T=20, Y=25. Count from the nearest one.

- **Ex 1:** Opposite of G. G=7, so $27-7 = 20 = \mathbf{T}$.
- **Ex 2:** BAD -> B(2) to 25=Y, A(1) to 26=Z, D(4) to 23=W = **YZW**.

Type 3 - Number coding (position values)

A=1 through Z=26. The code may be a sum, a list, or a product of positions.

- **Ex 1:** CAT = 24, since $3+1+20 = 24$. So DOG = $4+15+7 = \mathbf{26}$.
- **Ex 2:** PEN = $16-5-14$. So INK = **9-14-11**.

Type 4 - Coded sentence (substitution)

Find two sentences sharing exactly ONE word - their shared code is that word. Triangulate; never guess.

Worked example: "she is nice" = **pa lo ki**; "he is tall" = **lo mi da**; "she looks tall" = **pa zo da**.

- S1 and S2 share only **is**, and share only the code **lo** -> is = lo
- S1 and S3 share only **she**, code **pa** -> she = pa
- S2 and S3 share only **tall**, code **da** -> tall = da
- Leftover in S1: **ki** = nice

Type 5 - Symbol coding with a key

A table gives letter to symbol; substitute directly. **Ex:** A=@, B=#, C=\$ so **CAB = \$@#**.

Type 6 - Conditional coding

A symbol key **plus rules** that override it. Check the conditions first, apply the one that matches, then code the remaining letters normally.

Key: P=%, R=#, S=@, T=&, A=*, E=\$

(i) If the word **begins with a vowel**, code the first and last letters as the **last letter's code**.

(ii) If the word **ends with a vowel**, code the first and last letters as *****.

Ex 1: APT begins with vowel A -> rule (i) -> first and last both take T's code -> **& % &**

Ex 2: PSA ends with vowel A -> rule (ii) -> first and last both * -> *** @ ***

Practice - 12 questions

1. If CAT = DBU, then DOG = ?
2. If MOUSE = LNTRD, then TIGER = ?
3. If FRIEND = GTLISJ, then MOTHER = ?
4. What is the opposite letter of K?
5. If BAD = YZW, then CAB = ?
6. If CAT = 24, then DOG = ?
7. If PEN = 16-5-14, then INK = ?

For 8-10 use: "red apple sweet" = **ma pi ku**; "sweet mango juice" = **ku ta so**; "red juice cold" = **ma so li**.

8. What is the code for **juice**?
9. What is the code for **red**?
10. What is the code for **sweet**?

For 11-12 use the Type-6 key and rules above.

11. Code **APT**.
12. Code **PSA**.

Part 5 - Order & Ranking

1. **Total = (rank from left) + (rank from right) - 1** 12th from left, 18th from right $\rightarrow 12+18-1 = 29$
2. **Rank from the other end = Total - rank + 1** Class of 45, 20th from top $\rightarrow 45-20+1 = 26$ th
3. **Persons between = |difference of positions| - 1** 5th and 11th $\rightarrow 11-5-1 = 5$ persons
4. **Comparison ranking:** build a > chain, then read off.

Key phrasings: "heavier than only S" = 2nd lightest | "shorter than only Y" = 2nd tallest | "only two are taller than X" = X is 3rd tallest.

Worked example: Five boxes P, Q, R, S, T of different weights. Q is heavier than only S. R is lighter than P but heavier than T. "Q heavier than only S" makes S lightest and Q second. The rest are P, R, T with $T < R < P$. Lightest to heaviest: **S, Q, T, R, P**.

Practice - 8 questions

1. Ravi is 12th from the left and 18th from the right in a row. How many students in all?
2. In a class of 45, Sita ranks 20th from the top. What is her rank from the bottom?
3. In a row of 40, P is 15th from the left and Q is 20th from the left. How many sit between them?
4. Five boxes P, Q, R, S, T of different weights. Q is heavier than only S. R is lighter than P but heavier than T. Which box is the **heaviest**?
5. (Same set) Which box is the **3rd heaviest**?
6. In a row of 25 children, Amit is 11th from the left; Rohit is 7th to the right of Amit. What is Rohit's position from the **right** end?
7. A is taller than B but shorter than C. D is taller than A but shorter than C. Who is the **tallest**?
8. Meena is 16th from the top and 29th from the bottom. How many students are in the class?

Part 6 - Arithmetic

Six topics, each with the method, worked examples and practice. Answers in Part 7.

6.1 Profit & Loss

Term	Meaning
CP - Cost Price	what the seller paid
SP - Selling Price	what the seller sold it for
MP - Marked Price	the tag price before any discount

Profit = SP - CP | Loss = CP - SP

The percentage is **ALWAYS on CP**. Profit% = (Profit / CP) x 100. You earned it against what you *spent*, and you spent CP.

Forward (CP to SP): multiply. x% profit -> $x (1 + x/100)$. x% loss -> $x (1 - x/100)$.

Backward (SP to CP): DIVIDE. CP = SP / multiplier.

Discount is on MP. SP = MP x $(1 - d/100)$.

Why backward is division - the classic error, shown failing: CP 100 at 20% profit gives SP 120. Going back the wrong way, 120 - 20% of 120 = 96, not 100. The right way, 120 / 1.2 = **100**. The 20% was of CP, so you must undo a multiplication.

The three shortcuts that do the real work

A. Markup then discount: net% = $a - b - ab/100$. +40% then -25% -> 40-25-10 = +5% profit.

B. Same SP, one at +x%, one at -x%: ALWAYS a loss of $x^2/100$ %. Both at 960: the +20% article has CP 800 (profit 160); the -20% article has CP 1200 (loss 240). The loss sits on a bigger base, so net loss 80 on CP 2000 = 4%.

C. "CP of m articles = SP of n articles": profit% = $(m - n)/n \times 100$. CP of 15 = SP of 12 -> 3/12 = 25% profit.

Practice

1. CP = Rs 840, sold at 25% profit. Find SP.
2. SP = Rs 918 at a 15% loss. Find CP.
3. An article marked Rs 1,600 is sold at 25% discount, still giving 20% profit. Find CP.
4. Two articles are sold at Rs 960 each - 20% gain on one, 20% loss on the other. Overall profit or loss %?
5. The CP of 15 articles equals the SP of 12 articles. Find profit %.
6. Goods are marked 40% above cost and sold at 25% discount. Find profit %.
7. Selling at Rs 720 gives a 10% loss. What SP gives a 20% gain?
8. A trader gives 20% discount on MP and still earns 25% profit. If CP = Rs 600, find MP.

6.2 Simple & Compound Interest

SI = $P \times R \times T / 100$. Think in steps, not formula: **one year's interest = R% of P; multiply by T.**

Amount = $P + SI$.

Doubling: interest earned = 100% of P, so **$R \times T = 100$.**

Tripling: interest = 200%, so **$R \times T = 200$.** (Four times $\rightarrow$ 300.)

- **Ex 1:** Rs 6,000 at 10% for 4 years. 10% of 6000 = 600 per year, $\times 4 =$ **Rs 2,400.**
- **Ex 2:** A sum doubles in 8 years at SI. $R = 100/8 =$ **12.5%.**
- **Ex 3:** At 25%, time to triple: $25 \times T = 200$, so **$T = 8$ years.**

CI for 2 years - net rate = $2R + R^2/100$ %. 10% for 2 years = 21%. 20% for 2 years = 44%.

CI - SI over 2 years = $P \times (R/100)^2$. Works backwards too, to find P.

Amount under CI = $P (1 + R/100)^n$.

- **Ex 1:** CI on Rs 4,000 at 10% for 2 years $\rightarrow$ 21% of 4000 = **Rs 840.**
- **Ex 2:** CI-SI on Rs 8,000 at 5% for 2 years $\rightarrow 8000 \times (0.05)^2 =$ **Rs 20.**
- **Ex 3 (reverse):** CI-SI difference is Rs 40 at 10% for 2 years. $40 = P \times 0.01$, so **$P =$ Rs 4,000.**

Practice

1. Find the SI on Rs 6,000 at 8% per annum for 3 years.
2. A sum doubles in 8 years at SI. Find the rate.
3. A sum triples in 12 years at SI. Find the rate.
4. Find the CI on Rs 5,000 at 10% per annum for 2 years.
5. Find the difference between CI and SI on Rs 8,000 at 5% for 2 years.
6. The difference between CI and SI on a sum for 2 years at 10% is Rs 60. Find the sum.
7. Find the amount on Rs 4,000 at 10% compound interest for 2 years.
8. At what rate will Rs 2,500 amount to Rs 3,000 in 4 years at SI?
9. In how many years will a sum double at 12.5% SI?
10. Find the CI on Rs 1,000 at 20% per annum for 2 years.

6.3 Time & Work

The LCM method - kills all fractions. Total work = **LCM of the given days.** Each person's efficiency = total / their days. Add efficiencies to combine.

Two-person shortcut: together = **$a \times b / (a + b)$.**

Wages are shared in the ratio of efficiency (that is, of work actually done).

Man-days: $M_1 \times D_1 = M_2 \times D_2$ when the work is the same.

- **Ex 1:** A in 10 days, B in 15 days. LCM = 30. $A = 3/\text{day}$, $B = 2/\text{day}$, together $5/\text{day} \rightarrow 30/5 =$ **6 days.**
- **Ex 2:** A in 12, B in 6 $\rightarrow (12 \times 6)/(12+6) = 72/18 =$ **4 days.**
- **Ex 3 (alternate days):** A in 20, B in 30, working alternately from A. LCM = 60, $A = 3$, $B = 2$, so a pair of days does 5 units. $60/5 = 12$ pairs = **24 days.**

Practice

1. A does a job in 12 days and B in 18 days. How long together?
2. A does a job in 10 days and B in 15 days. How long together?
3. A and B together finish in 8 days; A alone in 12 days. How long does B alone take?
4. A is twice as efficient as B, and together they finish in 8 days. How long does A alone take?
5. 10 men complete a work in 15 days. How many days will 25 men take?
6. A does a work in 20 days, B in 30 days. They work on alternate days starting with A. In how many days is the work finished?
7. A, B and C can do a work in 10, 15 and 30 days. How long do all three take together?
8. A and B complete a work in 12 days and are paid Rs 600. A alone would take 20 days. Find A's share.

6.4 Partnership

Simple partnership (all invest for the same period): profit ratio = **ratio of capitals**.

Compound partnership (different periods): profit ratio = **ratio of (Capital x Time)**.

$$A : B = (C_A \times T_A) : (C_B \times T_B)$$

Working partner: take out the salary or commission **first**, then divide what is left in the capital-time ratio.

If capital **changes mid-year**, add each stretch separately: (first capital x months) + (new capital x months).

- **Ex 1:** A invests Rs 5,000 and B Rs 8,000 for the same period; profit Rs 2,600. Ratio 5:8 -> A = **Rs 1,000**, B = **Rs 1,600**.
- **Ex 2:** A invests Rs 6,000 for 8 months, B Rs 9,000 for 6 months. Ratio = 48,000 : 54,000 = **8 : 9**.
- **Ex 3 (mid-year change):** A puts in Rs 20,000, and after 6 months adds Rs 10,000; B puts in Rs 30,000 for the year. A = 20000x6 + 30000x6 = 3,00,000; B = 30000x12 = 3,60,000 -> **5 : 6**.

Practice

1. A invests Rs 4,000 and B Rs 6,000 for the same period. Profit is Rs 5,000. Find B's share.
2. A invests Rs 8,000 for 6 months and B Rs 6,000 for 8 months. Find the profit ratio.
3. A invests Rs 12,000 for a year, B Rs 15,000 for 8 months. Profit is Rs 11,000. Find A's share.
4. A, B and C invest Rs 3,000, Rs 4,000 and Rs 5,000 for the same period. Profit Rs 2,400. Find C's share.
5. A and B start with Rs 20,000 and Rs 30,000. After 6 months A adds Rs 10,000. Find the profit ratio at the end of the year.
6. A is a working partner and gets 10% of the profit as salary; the rest is divided in the capital ratio 2:3 (A:B). Total profit is Rs 11,000. Find A's total earning.

6.5 Mixtures & Alligation

The alligation cross: Cheaper : Dearer = **(Dearer - Mean) : (Mean - Cheaper)**.

Works for price, concentration, speed, marks - anything with **two sources and one average**.

Repeated replacement: if r litres are removed from v litres and replaced with water, n times, the pure liquid left = $v \times (1 - r/v)^n$.

- **Ex 1:** Mix rice at Rs 15/kg and Rs 20/kg to get Rs 18/kg. $(20-18) : (18-15) = 2 : 3$.
- **Ex 2:** Mix 30% acid and 60% acid to get 50%. $(60-50) : (50-30) = 10 : 20 = 1 : 2$.
- **Ex 3:** From 60 L of milk, 12 L is removed and replaced with water, twice. Milk left = $60 \times (1 - 12/60)^2 = 60 \times 0.64 = 38.4$ L.

Practice

1. In what ratio must rice at Rs 20/kg be mixed with rice at Rs 30/kg to get a mixture worth Rs 24/kg?
2. In what ratio must water (free) be mixed with milk costing Rs 32 per litre so that the mixture is worth Rs 24 per litre?
3. A 40 litre mixture has milk and water in the ratio 3:1. How much water must be added to make the ratio 3:2?
4. From a vessel holding 60 litres of milk, 12 litres is removed and replaced with water. This is done twice. How much milk remains?
5. Two vessels contain milk and water in the ratios 3:1 and 5:3. Equal quantities are mixed. Find the ratio in the final mixture.
6. The average age of a class of 30 students is 12 years. Some new students of average age 15 join, and the average becomes 13. How many joined?

6.6 Pipes & Cisterns

Identical to Time & Work, with signs. An inlet is **positive** work, an outlet or leak is **negative**.

Use the **LCM method**: tank capacity = LCM of the times; each pipe's rate = capacity / its time; add rates, subtracting for leaks.

If a tank fills in a hours alone but takes b hours with a leak, the **leak alone empties it in $ab / (b - a)$ hours**.

- **Ex 1:** A fills in 12 h, B in 18 h. LCM 36: A = 3, B = 2, net 5 $\rightarrow 36/5 = 7.2$ hours.
- **Ex 2:** A fills in 10 h, B empties in 15 h. LCM 30: A = +3, B = -2, net +1 $\rightarrow 30$ hours.
- **Ex 3:** A fills in 6 h but takes 8 h with a leak. LCM 24: A = 4, net = 3, so the leak = 1 $\rightarrow$ leak empties the full tank in **24 hours**.

Practice

1. Pipe A fills a tank in 12 hours and pipe B in 18 hours. How long if both are open?
2. Pipe A fills a tank in 10 hours; pipe B empties it in 15 hours. If both are open, how long to fill?
3. A pipe fills a tank in 6 hours, but with a leak it takes 8 hours. How long would the leak alone take to empty a full tank?
4. Two pipes fill a tank in 20 and 30 minutes. Both are opened together; after how long should the second be closed so that the tank fills in 18 minutes?
5. Three pipes fill a tank in 10, 15 and 30 hours. How long if all three are open?

6. A cistern has a leak that would empty it in 20 hours. An inlet admits 4 litres per minute, and with both working the cistern empties in 24 hours. Find the capacity of the cistern.

Part 7 - Answer keys

Seating sets A to G

SET A

Pos	1	2	3	4	5	6	7	8
Person	T	W	U	P	R	V	Q	S
Faces	N	S	N	N	S	N	S	S

A1. S A2. 2 (U and P) A3. P A4. 4 A5. R

SET B

Clockwise: **A(1), E(2), B(3), D(4), F(5), H(6), C(7), G(8)**. A = Doctor, F = Teacher, B = Lawyer, E = Engineer.

B1. G B2. F B3. 3 (D, F, H) B4. B B5. E

SET C

Column	1	2	3	4	5	6
Row 2 (North)	A	E	B	C	D	F
Row 1 (South)	T	Q	R	P	U	S

C1. B C2. A and F C3. 1 (D) C4. B C5. F

SET D

Clockwise: **J(1, corner), P(2, mid), K(3, corner), O(4, mid), N(5, corner), L(6, mid), M(7, corner), Q(8, mid)**. M likes Tea.

D1. M D2. Q D3. K D4. 2 (K and O) D5. J

SET E

Seat	1	2	3	4	5	6	7	8
Person	A	C	D	B	H	E	G	F
Faces	IN	IN	OUT	OUT	IN	IN	IN	OUT
Month	Apr	Dec	Feb	Jan	Sep	Jul	Oct	Mar

E1. F E2. G E3. 1 (B) E4. E E5. 5

SET F

Floor	Flat-1	Flat-2
5	D (White)	I
4	A	F (Blue)
3	G	C (Red)
2	B	J
1	E	H

F1. J F2. 1 (G) F3. F F4. J F5. D

SET G

Column	1	2	3	4	5
Row 2 (North)	P	S	R	Q	T
Row 1 (South)	D	C	B	E	A

Sports: P = Cricket, R = Tennis, E = Hockey, A = Football.

G1. S **G2.** R **G3.** D and A **G4.** S **G5.** A

Syllogism (20)

Q	1	2	3	4	5	6	7	8	9	10
Ans	e	a	a	a	a	c	c	d	a	a
Q	11	12	13	14	15	16	17	18	19	20
Ans	e	a	a	a	a	e	c	a	a	e

- **Q6, Q17:** All + Some, and the pair is Some $\leftrightarrow$ No $\rightarrow$ **Either/Or**.
- **Q7:** the pair is All $\leftrightarrow$ Some-not, the other complementary pair $\rightarrow$ **Either/Or**.
- **Q8:** Some + Some forces nothing, and "Some birds are black" vs "Some black are birds" is the **same claim twice**, not a negation pair $\rightarrow$ **Neither**.
- **Q1, Q11:** the conversion (All A are B $\rightarrow$ Some B are A) makes II true as well $\rightarrow$ **Both**.
- **Q5:** No + All is a forced mood: **Some C are not A**. II is not definite.
- **Q16, Q20:** the possibility conclusion holds *and* the definite one does $\rightarrow$ **Both**.

Coding-Decoding (12)

Q	1	2	3	4	5	6
Ans	EPH	SHFDQ	NQWLJX	P	XZY	26
Q	7	8	9	10	11	12
Ans	9-14-11	so	ma	ku	& % &	* @ *

- **Q3:** shift grows +1,+2,+3,+4,+5,+6: M+1=N, O+2=Q, T+3=W, H+4=L, E+5=J, R+6=X.
- **Q4:** K = 11, so 27-11 = 16 = P.
- **Q8-10:** S2 and S3 share only *juice* $\rightarrow$ **so**; S1 and S3 share only *red* $\rightarrow$ **ma**; S1 and S2 share only *sweet* $\rightarrow$ **ku**.
- **Q11:** APT begins with a vowel $\rightarrow$ first and last take T's code (&).
- **Q12:** PSA ends with a vowel $\rightarrow$ first and last become *.

Order & Ranking (8)

Q	1	2	3	4	5	6	7	8
Ans	29	26th	4	P	T	8th	C	44

- **Q1:** 12 + 18 - 1 = 29. **Q2:** 45 - 20 + 1 = 26th. **Q3:** 20 - 15 - 1 = 4.
- **Q4-5:** lightest to heaviest is S, Q, T, R, P. Heaviest = P; 3rd heaviest = T.

- **Q6:** Rohit is at $11 + 7 = 18$ from the left, so $25 - 18 + 1 = 8$ th from the right.
- **Q8:** $16 + 29 - 1 = 44$.

Arithmetic - answers with working

6.1 Profit & Loss

- 1. $840 \times 1.25 = \text{Rs } 1,050$
- 2. $918 / 0.85 = \text{Rs } 1,080$
- 3. $SP = 1600 \times 0.75 = 1200$; $CP = 1200 / 1.2 = \text{Rs } 1,000$
- 4. Shortcut B: $20^2/100 = 4\%$ **loss** (CPs 800 and 1200; total CP 2000, total SP 1920)
- 5. Shortcut C: $(15-12)/12 = 25\%$ **profit**
- 6. Shortcut A: $40 - 25 - 10 = 5\%$ **profit**
- 7. $CP = 720 / 0.9 = 800$; $SP = 800 \times 1.2 = \text{Rs } 960$
- 8. $SP = 600 \times 1.25 = 750$; $MP = 750 / 0.8 = \text{Rs } 937.50$

6.2 SI & CI

- 1. 8% of 6000 = 480/year, $\times 3 = \text{Rs } 1,440$
- 2. $R \times T = 100 \rightarrow 100/8 = 12.5\%$
- 3. $R \times T = 200 \rightarrow 200/12 = 16.67\%$
- 4. net 21% of 5000 = **Rs 1,050**
- 5. $8000 \times (0.05)^2 = \text{Rs } 20$
- 6. $60 = P \times (0.1)^2 = P \times 0.01 \rightarrow P = \text{Rs } 6,000$
- 7. $4000 \times 1.21 = \text{Rs } 4,840$
- 8. $SI = 500$; $500 = 2500 \times R \times 4/100 \rightarrow R = 5\%$
- 9. $12.5 \times T = 100 \rightarrow 8 \text{ years}$
- 10. net 44% of 1000 = **Rs 440**

6.3 Time & Work

- 1. LCM 36: $3 + 2 = 5 \rightarrow 36/5 = 7.2 \text{ days}$
- 2. LCM 30: $3 + 2 = 5 \rightarrow 6 \text{ days}$
- 3. LCM 24: together 3, A = 2, so B = 1 $\rightarrow 24 \text{ days}$
- 4. A : B = 2 : 1, total 3 units/day; A alone = $8 \times 3/2 = 12 \text{ days}$
- 5. $10 \times 15 = 150 \text{ man-days} \rightarrow 150/25 = 6 \text{ days}$
- 6. LCM 60: A = 3, B = 2, a pair of days = 5 units; $60/5 = 12 \text{ pairs} = 24 \text{ days}$
- 7. LCM 30: $3 + 2 + 1 = 6 \rightarrow 5 \text{ days}$
- 8. LCM 60: together 5, A = 3, B = 2, ratio 3:2 $\rightarrow$ A gets **Rs 360**

6.4 Partnership

- 1. $4000 : 6000 = 2 : 3 \rightarrow B = 5000 \times 3/5 = \text{Rs } 3,000$
- 2. $8000 \times 6 : 6000 \times 8 = 48000 : 48000 = 1 : 1$
- 3. $12000 \times 12 : 15000 \times 8 = 144000 : 120000 = 6 : 5 \rightarrow A = 11000 \times 6/11 = \text{Rs } 6,000$
- 4. $3 : 4 : 5 \rightarrow C = 2400 \times 5/12 = \text{Rs } 1,000$
- 5. $A = 20000 \times 6 + 30000 \times 6 = 3,00,000$; $B = 30000 \times 12 = 3,60,000 \rightarrow 5 : 6$

- **6.** Salary = 10% of 11000 = 1100. Remaining 9900 in 2:3 → A = 3960. A's total = **Rs 5,060**

6.5 Mixtures & Alligation

- **1.** $(30-24) : (24-20) = 6 : 4 = \mathbf{3 : 2}$
- **2.** Water costs 0: $(32-24) : (24-0) = 8 : 24 = \mathbf{1 : 3}$
- **3.** Milk 30, water 10. For 3:2 with milk 30, water must be 20 → add **10 litres**
- **4.** $60 \times (1 - 12/60)^2 = 60 \times 0.64 = \mathbf{38.4 \text{ litres}}$
- **5.** Take 8 units each: V1 gives milk 6, water 2; V2 gives milk 5, water 3 → 11 : 5 → **11 : 5**
- **6.** Alligation $(15-13) : (13-12) = 2 : 1$, so 30 : x = 2 : 1 → **15 students**

6.6 Pipes & Cisterns

- **1.** LCM 36: $3 + 2 = 5 \rightarrow 36/5 = \mathbf{7.2 \text{ hours}}$
- **2.** LCM 30: $+3 - 2 = +1 \rightarrow \mathbf{30 \text{ hours}}$
- **3.** LCM 24: pipe 4, net 3, leak = 1 → **24 hours** (or $ab/(b-a) = 6 \times 8/2 = 24$)
- **4.** LCM 60: A = 3, B = 2. A runs all 18 min = 54 units; the remaining 6 units come from B at 2/min → B runs 3 min, so **close it after 3 minutes**
- **5.** LCM 30: $3 + 2 + 1 = 6 \rightarrow \mathbf{5 \text{ hours}}$
- **6.** Inlet = 4 L/min = 240 L/h. $C/24 = C/20 - 240 \rightarrow 240 = C/120 \rightarrow \mathbf{28,800 \text{ litres}}$

Remember the one thing that decides your score. Speed has never been the problem - accuracy is. Every arrangement you build tends to be right; the marks leak in the final read-off. Slow down about 20 percent, say the count out loud, and you still finish with time to spare.